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Experimental Validation of Hydrogen Peroxide Injection for Heavy Oil Recovery: Insights from Medium-Scale Combustion Tube Test

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Abstract

Following the numerical and economic evaluation of hydrogen peroxide (H_2O_2) injection for heavy oil recovery, experimental validation is essential to reduce uncertainties and optimize field implementation. This study evaluates H_2O_2 through combustion tube (CT) testing, focusing on kinetic model validation, catalyst performance, and the effectiveness of H_2O_2 as a thermal EOR agent. The results refine numerical models and confirm the scalability of this technology for field applications, demonstrating its feasibility as an alternative to conventional thermal recovery methods.

A medium-pressure combustion tube (MPCT) test was conducted to replicate peroxide-driven oxidation and assess its impact on reservoir heating and oil displacement. The experiment was designed to evaluate multiple H_2O_2 injection cycles, catalyst efficiency, and the transition to *in situ* combustion (ISC). Experimental variables included core type, peroxide concentration, injection rates, and thermal front movement. The data obtained were used for history matching with numerical simulations, refining reaction kinetics, and improving process predictability in a field-scale hydrodynamic model.

The CT test validated the efficiency of H_2O_2 injection in generating controlled thermal effects, enhancing oil mobility, and initiating ISC. Catalyst-assisted peroxide decomposition resulted in localized heat generation, significantly reducing the required activation energy for oxidation reactions. Oil displacement efficiency was found to be highly dependent on permeability, peroxide concentration, and cyclic injection strategy. While higher peroxide concentrations accelerated thermal propagation, excessive oxygen production posed potential risks to well integrity. The test also demonstrated that a hybrid peroxide-oil injection cycle effectively maximized recovery by creating a uniform sweep and extending ISC front propagation. These experimental insights were later integrated into a field-scale model, confirming the feasibility of H_2O_2 – based thermal EOR under specific geological conditions.

This study provides the first comprehensive experimental validation of hydrogen H_2O_2 injection for heavy oil recovery using CT tests. By bridging the gap between theoretical modeling and field application, it enhances the understanding of peroxide decomposition kinetics, catalyst behavior, and process scalability.

The findings contribute new insights into peroxide-driven ISC initiation, offering a cost-effective and environmentally friendly alternative to traditional thermal EOR methods such as steam injection.

Introduction

The recovery of heavy oil poses persistent technical and economic challenges due to its high viscosity and poor flow characteristics under reservoir conditions (Fazlyeva et al., 2023; Moore, R. G. et al., 1996; A. T. Turta et al., 2007). Most technologies for producing high-viscosity oil rely on thermal enhanced oil recovery (EOR) methods, which reduce oil viscosity and improve mobility (Askarova, A. et al., 2020; A. G. Askarova et al., 2022; Lu et al., 2022; Mukhina et al., 2020; Popov et al., 2021; Wang et al., 2018). Key techniques include hot water flooding, which involves the injection of heated water to reduce oil viscosity (Goodyear et al., 1996; Kovscek et al., 2000; Sola, B.S. & Rashidi, F., 2008), steam injection, which similarly uses steam to lower viscosity and enhance oil mobility (Fadaei, H. et al., 2011; Gu et al., 2015); and *in situ* combustion (ISC), wherein a portion of the oil is burned to generate heat within the reservoir (Askarova et al., 2020; Ismail et al., 2016; Oskouei et al., 2010; A. Turta, 2012). ISC has also been investigated for *in situ* hydrogen production (A. Askarova et al., 2025; Hajdo et al., 1985; Kapadia, P.R. et al., 2009; Yuan et al., 2025), aligning with the growing interest in generating low-carbon fuels. Conventional thermal enhanced oil recovery (EOR) techniques, such as cyclic steam stimulation (CSS) (Hassanzadeh et al., 2009; J. G. Speight, 2019) and steam-assisted gravity drainage (SAGD) (Minakov et al., 2023; Pang et al., 2019), are widely applied but are associated with high energy consumption, significant water usage, and logistical constraints, particularly in remote or environmentally sensitive areas. These challenges often undermine their operational efficiency and economic viability, leading to increased interest in developing alternative thermal methods that are more sustainable and adaptable to diverse geological settings.

Recently, in in-reservoir heat generation via injection of thermochemical fluids has gained attention for enhancing well productivity (Anikin et al., 2022; A. G. Askarova et al., 2023). This technique involves injecting compounds that release energy upon decomposition within the formation. Among these, hydrogen peroxide (H_2O_2) has emerged as a promising EOR agent due to its strong oxidizing properties, ability to initiate ISC, application in thermobaric well treatments, and its capacity to reduce oil viscosity in cyclic injection schemes. H_2O_2 generates heat via catalytic decomposition, potentially enabling *in situ* reservoir heating without the need for surface steam generation (Takagi & and Ishigure, 1985). During the decomposition, H_2O_2 produces heat and oxygen according to the reaction:

$$2H_2O_2 \rightarrow 2H_2O + O_2 + \text{Heat}$$

This reaction can occurs at reservoir conditions in the presence of appropriate catalysts such as transition metal ions (e.g., Fe^{2+} or Mn^{2+}), enabling localized *in situ* heating and oxygen delivery without the requiring surface combustion or steam generation (Xing et al., 2018). The generated oxygen can then participate in oxidation reactions with in-place hydrocarbons, facilitating ISC initiation under controlled conditions. Unlike conventional ISC that relies on direct air injection, this method transitions to an enriched air ISC phase, where the reservoir is first pre-heated and partially oxidized by peroxide decomposition before being sustained by the injection of oxygen-enriched air (Tzanco et al., 1991; Yang et al., 2020). This staged process stabilizes the combustion front and improves thermal sweep efficiency.

A central aspect of the peroxide-based EOR process is the reaction kinetics governing peroxide decomposition and subsequent hydrocarbon oxidation. These reactions are temperature-dependent and influenced by a variety of factors, including peroxide concentration, catalyst type and availability, oil composition, and porous media properties. While peroxide decomposition follows pseudo-first-order kinetics in the presence of a catalyst, real reservoir systems introduce complexities such as heat losses, gas evolution, and competing side reactions. Similarly, heavy oil oxidation involves multiple reactions – including low-temperature oxidation (LTO), fuel deposition, and high-temperature combustion (HTO) –

each with distinct activation energies and reaction pathways. Accurate modeling and control of these kinetics are critical to ensure a stable and efficient thermal front that maximizes oil recovery while minimizing operational risks (Belgrave et al., 1993; Fazlyeva et al., 2023; Khakimova et al., 2020).

As previously mentioned, use of H_2O_2 injection as an oxidizing agent leads to enriched air injection (EAI). Normal air injection (NAI) and EAI differ fundamentally in their oxygen concentration and the resulting thermal and chemical behavior during ISC. While NAI uses atmospheric air with approximately 21% oxygen, EAI introduces higher oxygen concentrations—typically in the range of 30–40%—into the reservoir, significantly altering the combustion dynamics and enhancing thermal output and reaction kinetics. One primary impact of EAI is the increase in combustion front temperature and velocity. For example, combustion tube (CT) experiments by Rodriguez (Rodriguez & Mamora, 2005) showed that increasing oxygen concentration from 21% to 40% raised combustion front temperatures from ~450°C to 475°C and increased the front velocity from 13.4 cm/hr to 24.7 cm/hr. This acceleration enables faster heavy oil mobilization and reduces the time to production startup – from 3.3 hours under NAI conditions to just 1.8 hours with 40% EAI. These enhancements are linked to the effects of oxygen enrichment on ISC reaction kinetics, which involve a complex set of thermally activated reactions, including LTO, HTO, and thermal cracking or pyrolysis. Enriched air facilitates a faster and more efficient transition from LTO to HTO due to higher oxygen partial pressure and the elevated combustion zone temperatures.

EAI enhances ISC by accelerating reaction rates and lowering activation energies for high-temperature oxidation (HTO), enabling faster combustion. Doubling the oxygen concentration can increase reaction rates by three-to four times. Activation energies typically range from 20–30 kcal/mol for LTO and 35–45 kcal/mol for HTO, with enriched conditions potentially modifying these values due to altered reaction pathways and catalytic effects. EAI stabilizes the combustion front and reduces oxygen breakthrough by promoting more complete fuel oxidation. Although it accelerates combustion and advances oil production onset, it does not significantly boost ultimate recovery, which generally remains around 83% of original oil in place (OOIP). Thus, EAI improves ISC efficiency through faster kinetics and higher combustion intensity, it must be carefully managed to avoid risks such as excessive fuel consumption and combustion front instability at high oxygen concentrations.

The article by Yang (Yang et al., 2019) presented a detailed study on how EAI affects the performance and kinetics of ISC when combined with steam. Their findings indicate that the co-injection of steam and enriched air (95% O_2) can significantly enhance oil recovery, particularly in bitumen reservoirs that have already undergone thermal pre-treatment such as SAGD. The study highlighted, that EAI increases combustion front velocity and peak temperatures, especially when the process is initiated at high temperatures (125–220 °C). This results in faster initiation and stabilization of HTO reactions, which are more energy efficient and less prone to coke formation compared to LTO. The combustion front acted as a supplementary heat source supporting the steam front, which remains the dominate mechanism for bitumen displacement. Kinetic modeling in their study, based on Ramped Temperature Oxidation (RTO) data, incorporated ten reactions covering LTO, HTO, and pyrolysis. Key kinetic parameters, such as activation energies (ranging from 40–154 kJ/mol) and pre-exponential factors (spanning from 10^4 to 10^{14} day^{-1}) were tuned to match CT test data. Enriched air was found to not only accelerate fuel consumption but also shift the dominant fuel from coke to lighter hydrocarbons, which react more rapidly with oxygen. Consequently, EAI enables a more efficient ISC process by reducing coke deposition and enhancing thermal front propagation. Overall, the co-injection strategy led to ultimate recovery factors of up to 90% of OOIP in laboratory settings, while providing valuable insights for field-scale optimisation.

A critical component of the peroxide-based EOR method is the effective initiation of peroxide decomposition under reservoir conditions, which typically requires a catalyst (Anikin et al., 2022; A. G. Askarova et al., 2023). In this study, several catalytic systems were evaluated to trigger the thermal reaction at reservoir temperature ranges (60–120°C), including aqueous potassium permanganate ($KMnO^4$), manganese dioxide (MnO_2), and iron (III) oxide (Fe_2O^3) nanoparticle suspensions. Among them, aqueous $KMnO^4$ (5–6% solution) was selected as the most cost-effective and operationally practical catalyst for field implementation due to its simplicity of injection and reliable performance. Upon interaction with in-reservoir hydrocarbons, $KMnO^4$ undergoes a redox reaction forming MnO_2 on the oil–water interface, which then serves as a solid-phase catalyst for H_2O_2 decomposition. This catalytic layer initiates oxygen and heat generation upon contact with H_2O_2 effectively triggering thermal processes and pre-ignition reactions in situ. While MnO_2 enables multiple activation cycles, $KMnO^4$ and Fe_2O^3-based systems act as single-use catalysts. According to laboratory validation a 48-hour soaking period of the catalyst in the core model prior to peroxide injection was essential for forming a sufficient reactive surface area (Lin & Gurol, 1998). This method ensures localized heat release and reliable combustion onset, contributing to greater control of the thermal front and enhanced oil displacement. The catalyst selection and delivery protocol were experimentally confirmed using intact core samples from the B2 reservoir of the Maiorovskoye field, making this approach viable for subsequent field-scale pilot testing.

Previous studies on H_2O_2 injection have primarily focused on numerical simulations to evaluate its feasibility, including its thermal output, and economic viability. In the work by (Askarova et.al., 2023) H_2O_2 injection experiments confirmed two heating stages: peroxide decomposition and oil oxidation leading to combustion. Sensitivity studies were performed, and the target reservoir was selected for pilot testing based on economic analysis. However, experimental validation at a laboratory scale remains essential to reduce uncertainties in reaction modeling, catalyst delivery, and combustion behavior, particularly under enriched air conditions

This study addresses this gap by conducting a MPCT experiment designed to replicate the physical and chemical processes occurring during peroxide injection, catalytic decomposition, and the transition to enriched air ISC. The objectives were to quantify the thermal and recovery performance of the process, validate kinetic models used in prior simulations, and assess the operational conditions required for successful field-scale implementation. Specifically, the primary goals were to (1) assess the thermal efficiency and oil displacement performance of H_2O_2 injection, (2) evaluate the role of catalysts in controlling decomposition kinetics, and (3) refine predictive models for field-scale application.

By integrating experimental observations with numerical history matching, this study offers new insights into peroxide-driven oxidation kinetics and the practicalities of using H_2O_2 as a precursor to ISC. The findings contribute to the growing body of research supporting peroxide-based EOR as a cost-effective and environmentally compatible alternative to steam-based methods, particularly in reservoirs where water availability, infrastructure, or depth constraints limit the use of conventional thermal technologies.

Methods and materials

The MPCT test simulated the heavy oil recovery with a new thermal technology based on injection and in-situ H_2O_2 decomposition. It was conducted following a standard CT methodology typically used for steam or hot water flooding experiments and ISC tests (A. G. Askarova et al., 2022; Popov et al., 2021). To closely replicate reservoir conditions, consolidated core samples from the target reservoir and wellhead crude oil samples were used. This section describes the preparation of the core model, the CT setup, additional equipment, and the experimental procedure.

Rock samples

The target oil field locates in Volga-Ural oil and gas region. The formation, at depth of 1600-1700 meters, is represented by sandstone rocks with high porosity and permeability. At this depth and at reservoir pressure of 15 MPa, steam thermal recovery operates as hot water injection. The mineralogical composition consists of up to 94,6 % silica, approximately 5% clay minerals, and less than 1% of carbonates. The low carbonate content allows for the full utilization peroxide decomposition potential in the formation, as hydrothermal reactions from carbonate decomposition are absent.

As mentioned above, the laboratory experiment was carried out on a consolidated core. The composite core model for the CT consisted of cylindrical samples with a diameter of 50 mm and a total length of 961 mm. To prepare the core model, 25 cylindrical samples were drilled from full-size core sections and cleaned using a CO_2 extractor and Soxhlet apparatus with toluene until the solvent was decolorized. After extraction, the core material was dried in an oven at 70°C. During extraction, three samples collapsed due to weak mineral matrix consolidation and removal of the binding heavy oil. The remaining samples were trimmed to create a straight column of the required length.

The cores were scanned by CT to detect of heterogeneities, fractures, or cavities. If any significant defects were identified, the stacking order was adjusted accordingly. The assembled column was then placed into the core holder and cemented using a chemically and thermal resistant composition. Additionally, five smaller cores (30 mm) were drilled along the length of the full-size section to determine reservoir properties using a helium permeameter. Figures 1 and 2 show core samples from target formation that were selected and delivered to the laboratory for research.

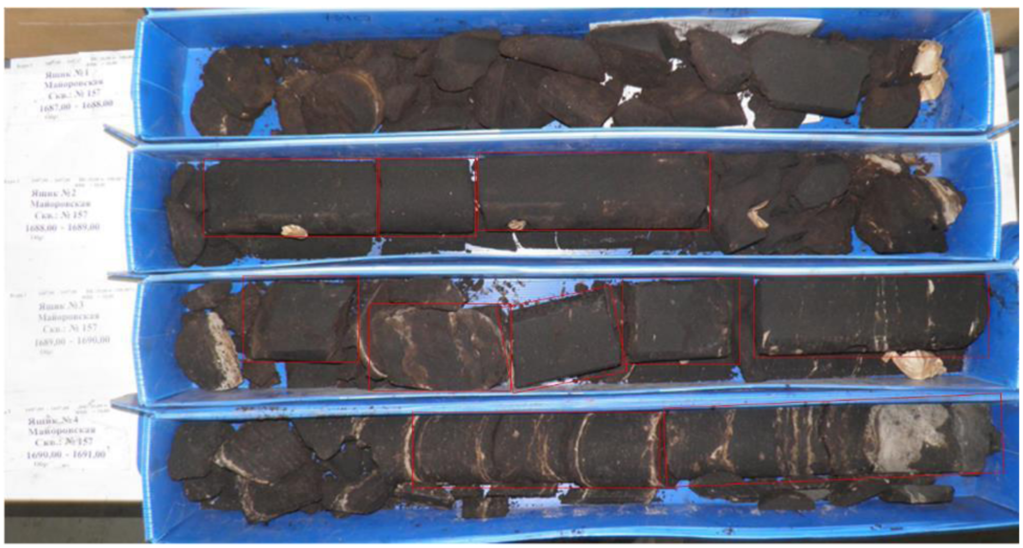


Figure 1—Full-size cores of target sandstone reservoir

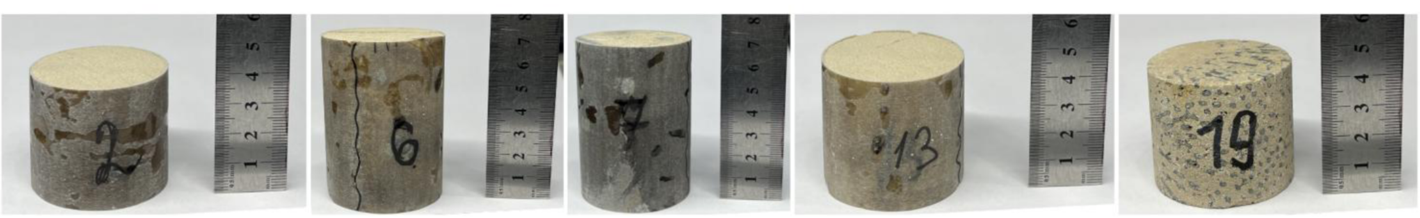


Figure 2—Some cleaned cylindrical core samples of the target formation (2, 6, 7, 13, 19) and artificial samples (19) used in the experiment

The scheme of the core model used in the experiment is shown in Figure 3. At both sides of the column, additional samples, marked as AS composed of dolomite rock, were used to replace the destroyed sandstone cylinders. The procedure of sealing the core model into the stainless-steel core holder followed the methodology described earlier in (A. Askarova et al., 2020; A. G. Askarova et al., 2022). The final total length of the core column was 961 mm. Reservoir properties were measured for five of the samples, yielding an average open porosity of 27.0 % and an absolute permeability of 1.77 D.

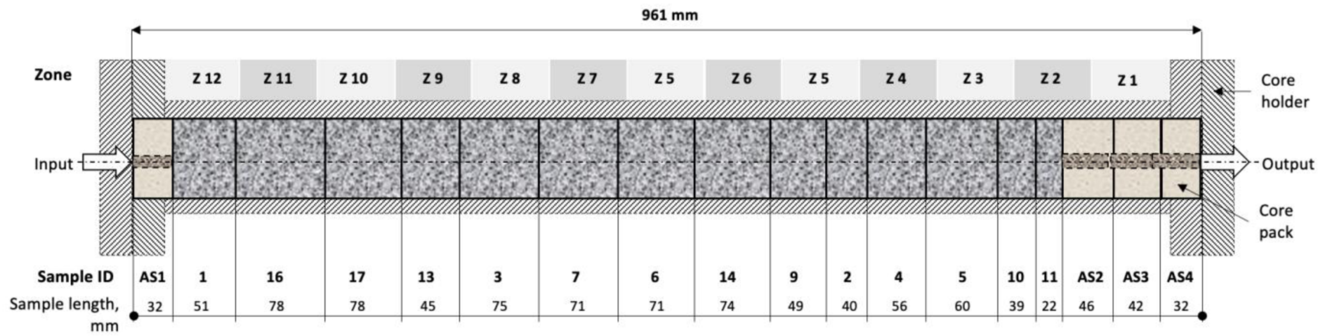


Figure 3—The scheme of the core model

Oil samples

For the test, 3.2 L of the dehydrated and purified dead oil from target oil field were prepared. Temperature-dependent density and viscosity were measured using a Mettler Toledo Excellence D4 density meter according to ASTM D4052 (ASTM D4052, 2018) and a Physica MCR 302 rheometer following ASTM D4440-15 (ASTM D4440-15, 2015). The results are presented in Table 1 and Table 2.

Table 1—Oil density

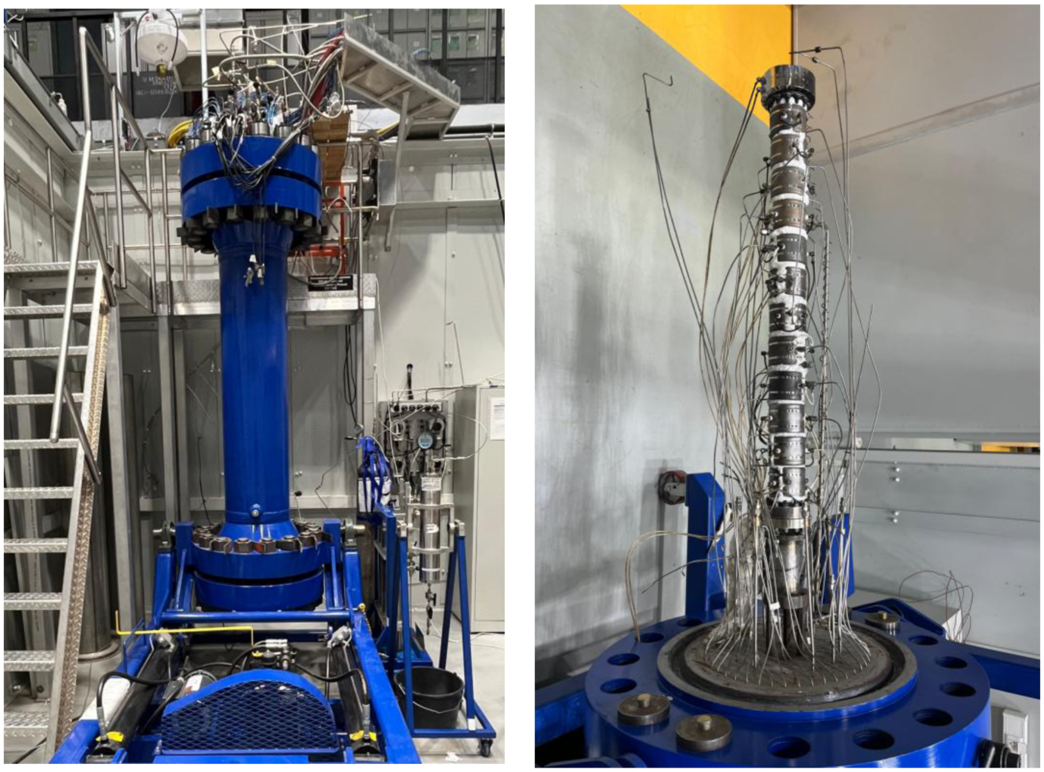
Sample	Density, g/cm³				
	20°C	25°C	30°C	35°C	40°C
Dehydrated oil well #3	0.939	0.936	0.932	0.928	0.923

Table 2—Oil viscosity

Sample	Viscosity, mPa·s							
	25°C	30°C	35°C	40°C	45°C	50°C	60°C	70°C
Dehydrated oil well #3	240.50	170.25	128.20	99.20	78.09	62.53	43.20	31.43

Experimental procedure

The experiment was conducted using the MPCT shown in Figure 4. A comprehensive description of the experimental setup is available in previous studies (A. G. Askarova et al., 2022; Popov et al., 2021). A distinctive feature of this experiment was the use of a composite core model assembled from cylindrical sandstone samples with a diameter of 50 mm, enabling conditions that closely replicate actual reservoir conditions. The core model was placed within a custom-designed thin-wall stainless steel core holder approximately one meter in length, equipped with 12 pairs of internal and wall thermocouples for detailed thermal monitoring.



(a) (b)

Figure 4—MPCT: a) assembled CT, b) core holder with connected internal and wall TC.

The experiment simulates one cycle of H_2O_2 injection technology and consisted of the following stages:

1. Introduction of a 6% mass. aqueous $KMnO^4$ solution as a precursor, followed by a 48-hour soak period to create the MnO_2 catalyst. Reservoir pressure and initial temperature were established in the model.
2. Injection of H_2O_2 with monitoring of thermal effect — specifically decomposition of H_2O_2 followed by ISC and collection of the produced fluids and gas.
3. Purging with water to remove unreacted oxygen, residual H_2O_2, and combustion gas.

Temperatures within the core model and on the wall for 12 zones, along with confining, inlet and outlet pressures, as well as volumetric flow rates of injected water, oil, H_2O_2 solution, all logged by the data acquisition system. Gas composition was analyzed using two gas chromatographs at 25 minutes interval.

Porosity and permeability of the core model were determined during a water filtration test which was conducted after water saturation. A summary of the core model description and experimental conditions is presented in Table 3.

Table 3—Experimental conditions

Parameter	Value	Note
Reservoir/pore pressure, MPa	15.0-15.3	corresponds to the pressure of the target reservoir
Lithostatic/Confining pressure, MPa	18.0	–
Type of core model	cylindrical cores	cleaned sandstone cores
Sandstone cores length, mm	810	zone 2-15
Additional cores length, mm	151	zones 1, 16, 17, 18
Total length of the core model, mm	961	–

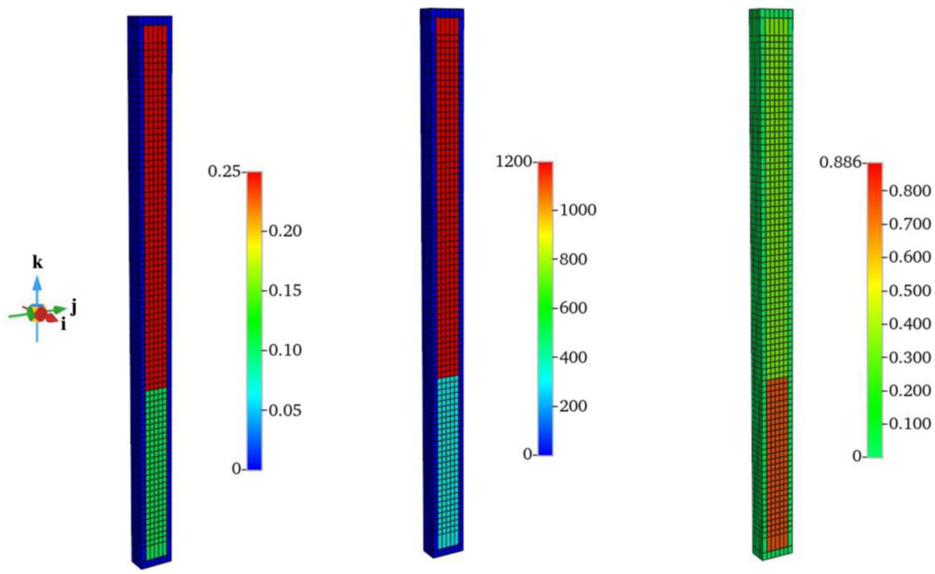
Parameter	Value	Note
Diameter of the core model, mm	50	–
Water permeability of the model, D	1.7	–
Average porosity of the model, units.	0.26	–
Oil viscosity, mPa·s (at 25 °C)	241	–
Oil density, g/cm^3 (at 25 °C)	0.936	–
Initial oil saturation, units	0.35	after catalyst injection
The precursor/catalyst	$KMnO_4$ 6% aqueous solution	
H_2O_2	50% water solution	–
Initial temperature of the model, °C	26	–
Injection rate H_2O_2, ml/min -average -max	4 12	–
Injection rate H_2O, ml/min	5-10	–

The input data for history matching of the experiment were prepared based on laboratory results and core model parameters. The initial dataset for numerical modeling included:

- geometric dimensions and material properties of CT and core holder;
- oil properties for different stages of the experiment, including fractional composition of maltenes and asphaltenes and molecular weight;
- porosity and permeability of the core pack;
- properties of the H_2O_2 solution;
- experimental results: initial oil and water saturations, injection rates, temperature profiles, masses of produced water and oil, produced gas composition and volume, recovery factor (RF), combustion front velocity.

Numerical simulation

For the numerical simulation of the experiment, the thermal-hydrodynamic simulator CMG STARS was employed. Development of the hydrodynamic model (HDM) commenced with the constructing the geometry of the CT core holder using a Cartesian grid (see Figure 5). This approach allowed for accurate reproduction of the geometric parameters of the entire setup, including the core pack properties.



(a) (b) (c)

Figure 5—Cartesian grid of simplified model with a more uniform mesh: a) porosity, b) permeability, c) oil saturation.

A Cartesian grid was selected due to its ability to provide a structured discretization in the (i, j, k) directions, enabling more precise representation of fluid and gas phase hydrodynamics, as well as reaction kinetics. Previous modeling experience has shown that Cartesian grids are particularly effective for simulating ISC processes because of its geometric simplicity and computational efficiency. In contrast, radial grids, despite better geometric conformity in cylindrical systems, result in uneven cell distribution, leading to increased computation time when modeling combustion front propagation (Lehrstuhl, 2024).

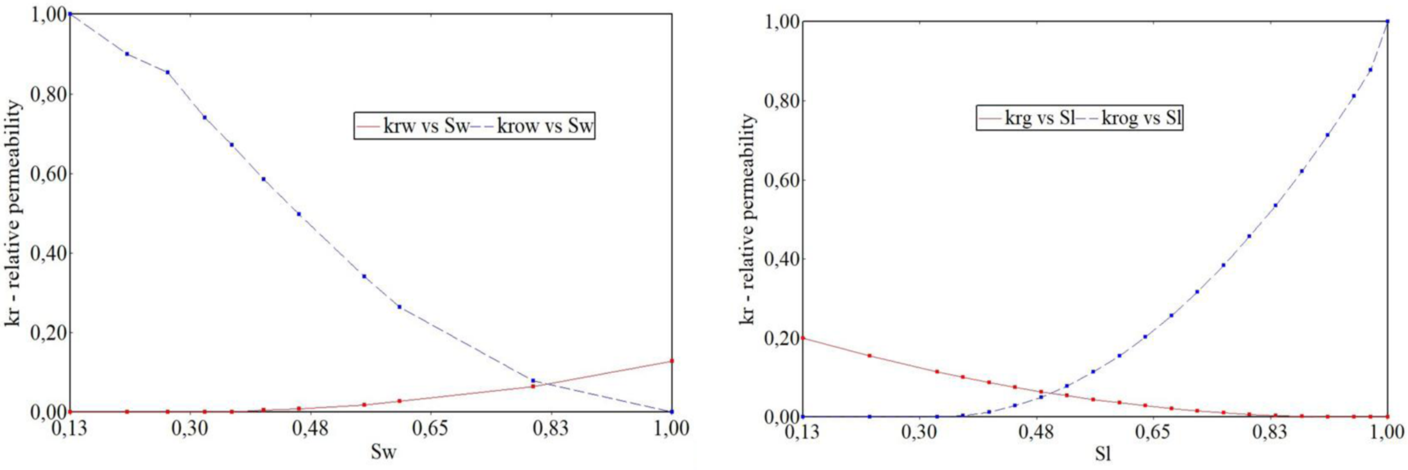
Laboratory results indicated a reduction in permeability in zones 12–9 near the injection end of the core model during oil and water saturation and catalyst injection. To replicate this in the numerical model, reduced permeability values were assigned to these zones (see Figure 5), ensuring accurate representation of experimental initial conditions. In the model, H_2O_2 was injected uniformly across the cross-section of the core holder at a height of 31 cm from the bottom. The core holder wall thickness was set to 1 cm to facilitate the creation of uniform computational grid and accelerate calculations, an assumption justified by the rapid progression of the thermal and chemical processes under study.

The constructed numerical model consists of seven cells in the horizontal direction (i), 7seven cells in the azimuthal direction (j), and 88 cells in the vertical direction (k). The outer cells corresponding to the model walls have dimensions of 1 cm in the i and j directions, and 1.5 cm at the outer boundaries in the k direction. Porous medium cells have a vertical size of 1.09 cm. The parameters of the numerical model reproduce the average properties of the laboratory core model and are presented in Table 4. The relative permeability curves were selected based on the target formation, presented in Figure 6.

Table 4—Thermal properties of the core, water, oil, a gas phase used in the hydrodynamic model.

Thermal conductivity of rock, W/(m·K)	1.63
Volumetric heat capacity of rock (temperature dependency), MJ/(m^3·K)	$1.86 + 3.3 \times 10^{-3} T$
Thermal conductivity of water*, W/(m·K)	0.58
Thermal conductivity of oil*, W/(m·K)	0.14
Thermal conductivity of gas phase*, W/(m·K)	0.05

* values correspond to simulator defaults.



(a) (b)

Figure 6—Relative permeability curves: (a) oil-water system, (b) gas-oil system

The thermal properties of the rock were defined by analogy with sandstone reservoirs saturated with heavy oil, as reported by (Chekhonin et al., 2020), and are listed in Table 4. The fluid model consisted of 13 pseudocomponents, with molecular weights presented in Table 5.

Table 5—Molecular weights of the pseudocomponents used in the hydrodynamic model

Component	H_2O_2	H_2O	H_2S	$C_2H_6\_NC_4$	NC_5toFC_7	MALT	ASPH
Molar mass, g/mole	34	18	34	41.9	77	300	1100
Component	CH_4	N_2	CO_2	O_2	CO	COKE	
Molar mass, g/mole	16	28	44	32	28	12.5	

Boundary conditions and injection parameters in the numerical model matched the experimental setup. A 50% H_2O_2 solution was injected at an average rate of 3 ml/min at an initial temperature of 22 °C, followed by a water flushing stage to displace residual gases.

The kinetic reaction model was based on the model developed by J. Belgrave (Belgrave et al., 1993; Moore et al., 1992) incorporating four oil oxidation reactions: thermal cracking, two LTO reactions and one HTO oxidation reaction. For this study, an additional reaction for H_2O_2 decomposition was incorporated, with adapted kinetic parameters presented in the Numerical results section. Initial oil oxidation parameters were sourced from (Maksakov et al., 2021), corresponding to the target field. The model accurately reproduces the experimental setup, with minor adjustments. Its advantages include computational efficiency due to simplification while retaining accuracy in representing crushed core properties and thermal behavior. However, it uses averaged injection values and neglect heat losses through the core holder walls.

Results

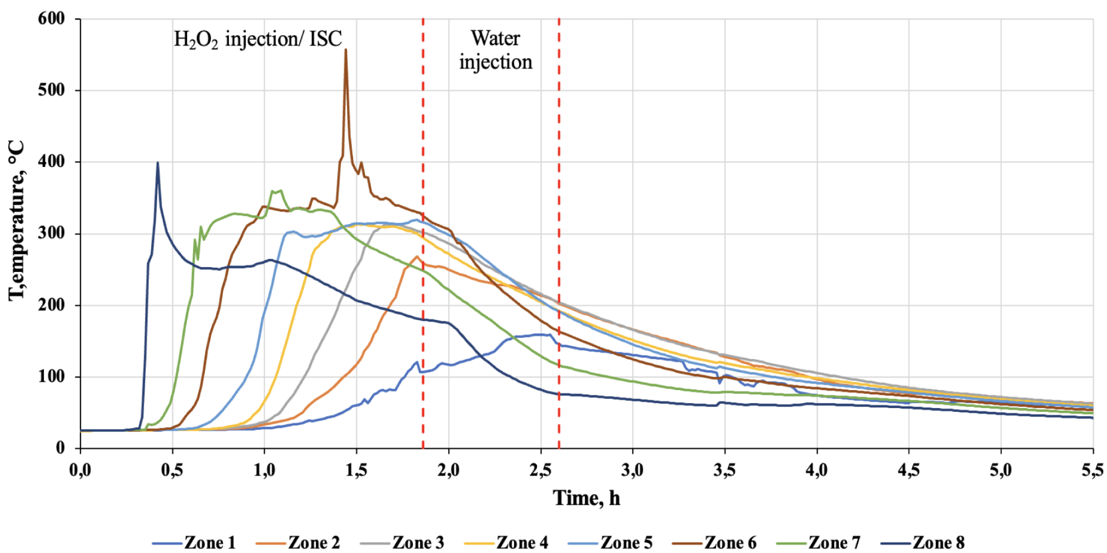
Experimental results

The main phase of the experiment was successfully conducted in accordance with the established methodology. However, a modification was required during the preparation stage. Plugging occurred in injection zones 9–12 during catalyst injection, likely caused by fines from sandstone transported with heavy oil. To address this issue, the injection line was reconnected to a mid-length port, thereby excluding the bottom low-permeable zone and reducing the active model length to 640 mm (Figure 5). A single H_2O_2

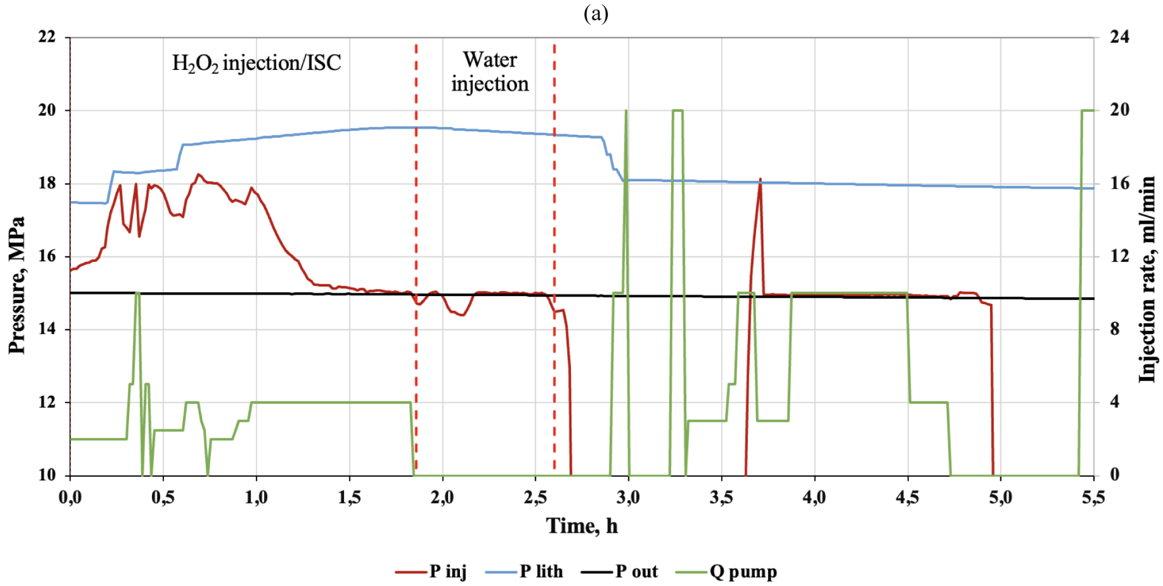
injection cycle was performed. Table 6 summarizes the main experimental events. Temperature profiles across zones 1–8 are shown in Figures 7a, while pressure profiles are presented in Figure 7b. Numerical modeling and history matching were carried out using data from this H_2O_2 injection cycle.

Table 6—Main events during the experiment

	Event	Time, h		Volume, ml	Pore volume, units
		start	end		
1	Pressure increase	-0.67	0		
2	H_2O_2 injection and ISC	0	1.86	377	0.79
3	Start of H_2O_2 decomposition and combustion	0.42			
4	Water Injection (Zones 8-1)	1.91	2.6	429	0.90



(a)



(b)

Figure 7—(a) Temperature profiles for zones 1-8; (b) pressure graphs at the model inlet (red line), outlet (black line), and confining pressure (blue line).

Peroxide decomposition begins 19 minutes after the start of injection at a rate of 2 ml/min; thereafter, the average injection rate of H_2O_2 varied, as shown in Figure 7b. The temperature in the injection zone 8 reached 400°C before decreasing and stabilizing around 250°C. Oxygen release and heat generation from peroxide decomposition promptly initiated oil oxidation reactions. The combustion front propagated uniformly with continued H_2O_2 injection injection towards zones 1–2 at an average velocity of 0.33 m/h. Injection was stopped once the combustion front reached these zones. Temperature peaks along the core pack ranged from 310–567°C, with the highest temperature observed in zone 6 (567°C), likely associated with combustion within a cavity around the internal thermocouple.

Simultaneous peroxide decomposition and ISC caused a pressure rise during the first half of the treatment as produced gases established stable conductivity through the model. The injection rate was later reduced to maintain pore pressure below the allowable limit for the CT. The average injection pressure increase (Figure 7b) was 2.6 MPa, approximately 17% above pore pressure, during this stage. In subsequent stages, the pressure drops across the model stabilized at around 0.1 MPa. The observed pressure behavior during peroxide decomposition initiation and ISC onset indicates that H_2O_2 injection rates will need to be adjusted during field-scale application. Therefore, an appropriate pump must be selected to provide planned injection rates while responding to downhole pressure fluctuations.

The observed ISC regime corresponded to wet or super-wet combustion, with a water-to-air ratio of 1 L/m^3 and an average temperature of 320°C. The calculated fuel requirement was 21.8 kg/m^3, corresponding to approximately 12% of the initial oil volume. Based on material balance calculations, the recovery factor was 0.66 for the entire core model and approximately 0.81 for zones thermally treated with H_2O_2 and ISC. Produced oil exhibited variable properties, with density ranging from 0.936 to 0.973 g/cm^3 and viscosity from 241 to 3074 mPa·s (at 25°C and 1 atm). Despite this variability, high temperatures maintained oil mobility. Asphaltene content increased due to oxidation and the presence of manganese residues. Nearly complete oxygen utilization (98%) was achieved. Additional findings included full consumption of the initiator after treatment. Concentration of the manganese in the rocks measured with XRF after treatment reduced by 2.3 times from 11 400 to 4 850 ppm. The manganese displacement during filtration observed also. The fluid sample consists of Mn with 1 020 ppm concentration. The oil and water cleaning procedure of the sandstone cores caused partial core destruction and fines migration within pores, likely contributing to the observed plugging. Samples from these zones were destroyed during unpacking, preventing further structural analysis. This effect requires additional investigation to develop mitigation strategies for preventing wellbore plugging during precursor injection in field applications.

Numerical results

This section presents the quantitative results derived from the adaptation of the numerical model, with a focus on key computational outcomes and observed trends supported by experimental and fluid property data. The adaptation process commenced with calibration of the reaction kinetics to ensure consistency with the experimentally characterized fluid model. Calculations incorporated quantitative fluid properties, including measured gas-to-produced oil ratios, hydrogen-to-carbon (H/C) ratios, and compositional analyses, which are critical for accurately representing oxidation and combustion reactions.

The adapted kinetic model, summarized in Table 8, was developed by calculating the stoichiometric coefficients for each reaction based on the molecular weights and fractional compositions of the fluid components. This ensured mass balance consistency and improved the predictive capability of the model for simulating thermal decomposition, oxidation kinetics, and resulting fluid behavior under reservoir conditions.

Table 7—Kinetic reaction model

№	Reactions
1	Thermal cracking
	ASPH → 2.786666667 MALT + 2.5 CO_2 + 2.75 CH_4 + 8.801408225 COKE
2	LTO 1
	MALT + 2.53125 O_2 → 0.346363636 ASPH
3	LTO 2
	ASPH + 7.5625 O_2 → 107.3771803 COKE
4	HTO
	COKE + 1.124500 O_2 → CO_2 + 0.249000 H_2O
5	H_2O_2 decomposition
	H_2O_2 → H_2O + 0.5 O_2

Table 8—Kinetic parameters of the reactions used in the numerical model

№	Activation energy E_a, J/mole	Frequency factor A, h-1	Frequency factor* A, h-1	Enthalpy H, J/mole
1	181041	2.7597×10^{16}	2.7597×10^{16}	0
2	86730	2.4×10^{9}	2.4×10^{9}	546740
3	185600	2.4×10^{9}	2.4×10^{9}	3141300
4	34763	2.4×10^{8}	2.4×10^{9}	470810
5	55000	1×10^{12}	8×10^{9}	98000

* adapted values

The values of the determined kinetic parameters for the reactions used in the numerical model are presented in Table 8. The adapted results differ from the literature data (Belgrave et al., 1993; Fazlyeva et al., 2023; Khakimova et al., 2020), particularly in the frequency factors of the fourth and fifth reactions, as well as in the reaction order of the third reaction (refer to the Table 7).

To account for the catalytic effect observed in the experiment, a reduced activation energy for the H_2O_2 decomposition reaction—derived from prior experimental studies—was applied. Additionally, the pre-exponential factors for both H_2O_2 decomposition and high-temperature oxidation reactions were used as tuning parameters to regulate reaction rates and extents.

Model calibration revealed that LTO reaction No. 2, which involves the conversion of asphaltenes into coke, significantly influences the overall process, particularly affecting subsequent HTO behavior. For this reaction, the following reaction orders (degrees of freedom) were specified: 3 for the O_2 component and 1 for the ASPH component. Adjusted relative phase permeabilities were also employed in the model.

Calibration of the hydrodynamic model (HDM) for the H_2O_2 injection experiment into the core model was performed using experimental temperature profiles and the cumulative volume of carbon dioxide produced, which served as a key indicator of the HTO process. During the experiment, the oxygen volume fraction at the model outlet ranged from 0.3% to 5% after HTO initiation, which is relatively low. The corresponding oxygen fraction in the HDM was also less than 1% of the total gas volume (approximately 0.1 liters); therefore, no separate adjustment for oxygen concentration was required.

Figure 8 presents the temperature profiles calculated by the HDM alongside experimental data for six CT zones where H_2O_2 decomposition and HTO reactions occurred. Decomposition of H_2O_2 and initiation of HTO began in injection zone 8, with the HTO front subsequently propagating through zones 7 to 3. Reagent injection was terminated once the front reached zone 3.

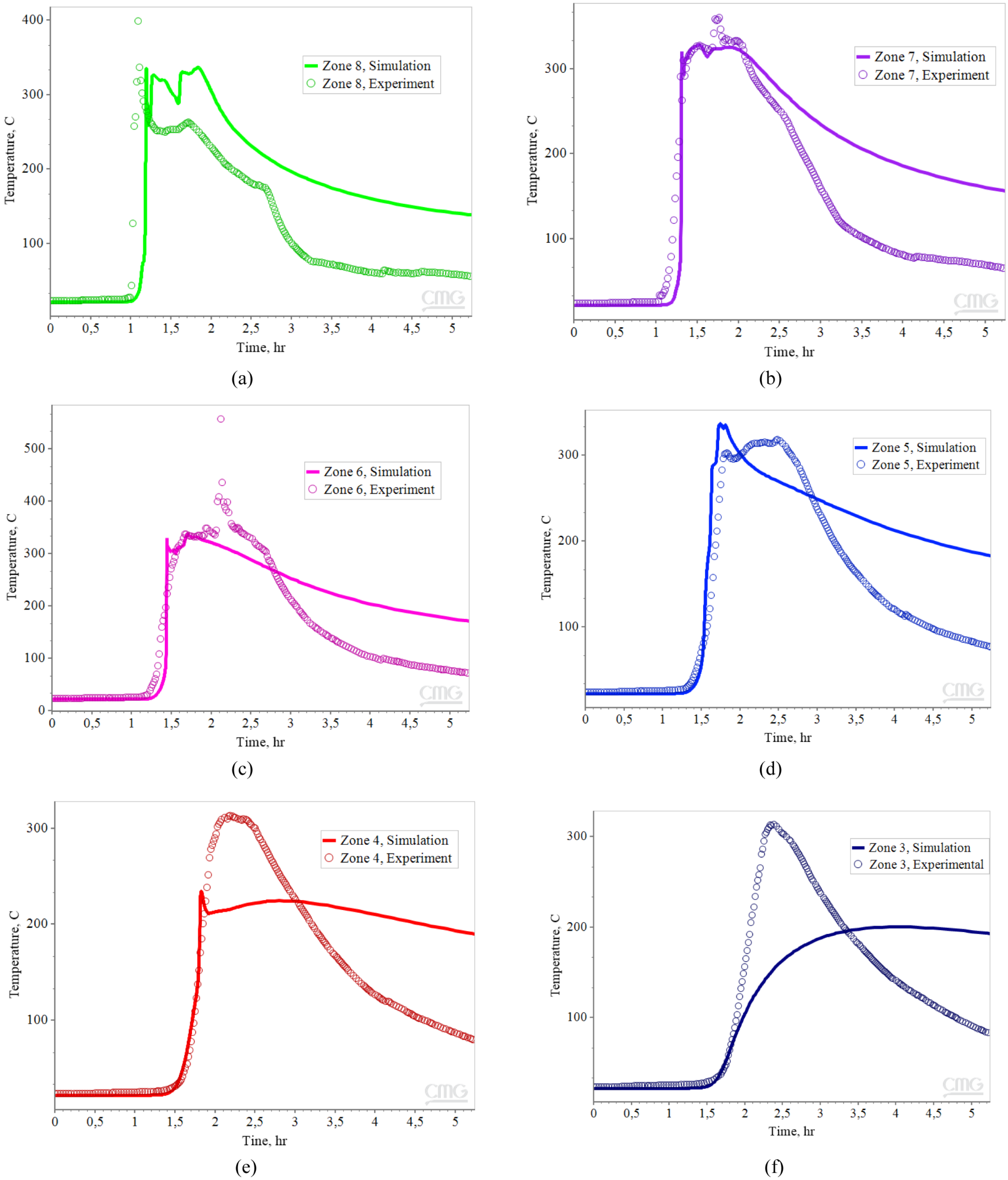


Figure 8—Experimental and numerical temperature profiles: (a) Zone 8; (b) Zone 7; (c) Zone 6; (d) Zone 5; (e) Zone 4; (f) Zone 3

The temperature profiles for zones 7, 6, and 5 show good agreement between the numerical model and experimental data in terms of both curve shape and peak temperature values. In contrast, for zones 4 and 3 (Figure 8d, e), discrepancies are observed in the local maximum temperatures and the shape of the temperature curves. This divergence may be attributed to the variable volumetric rate of H_2O_2 injection in the experiment, while the numerical model employed a constant average injection rate. This difference likely affected the dynamics of temperature front propagation in the last two zones of the model.

In zone 8, where both H_2O_2 decomposition and oxidation reactions occurred simultaneously, differences in the average temperature reached were also noted. A possible explanation for this discrepancy is the heterogeneous initiation of oil oxidation reactions in the laboratory sample, whereas the numerical model assumes a more uniform reaction distribution, influencing the total heat generated.

One of the most critical parameters of combustion is the propagation velocity of the temperature front, reflected in the zones with rapid temperature rise and peaks shown in Figure 8a–e. The model results indicate that the timing of front propagation through the zones closely matches the temperature rise observed in the experiment, accurately reproducing the combustion front movement. Furthermore, the temperature peak heights in the model represent an averaged picture of the experiment and correspond to the high-moisture oil combustion regime.

The distribution of temperature and oil saturation over time is presented in Figure 9. In the temperature graph (Figure 9a), the blue dot in zone 8 (2-5 hours) indicates the beginning of water injection to displace the peroxide rim and remove residual reagents, resulting in local cooling. Similarly, the oil saturation graph (Figure 9b) shows a stabilization of values during this period, associated with the completion of the active combustion phase and oil displacement.

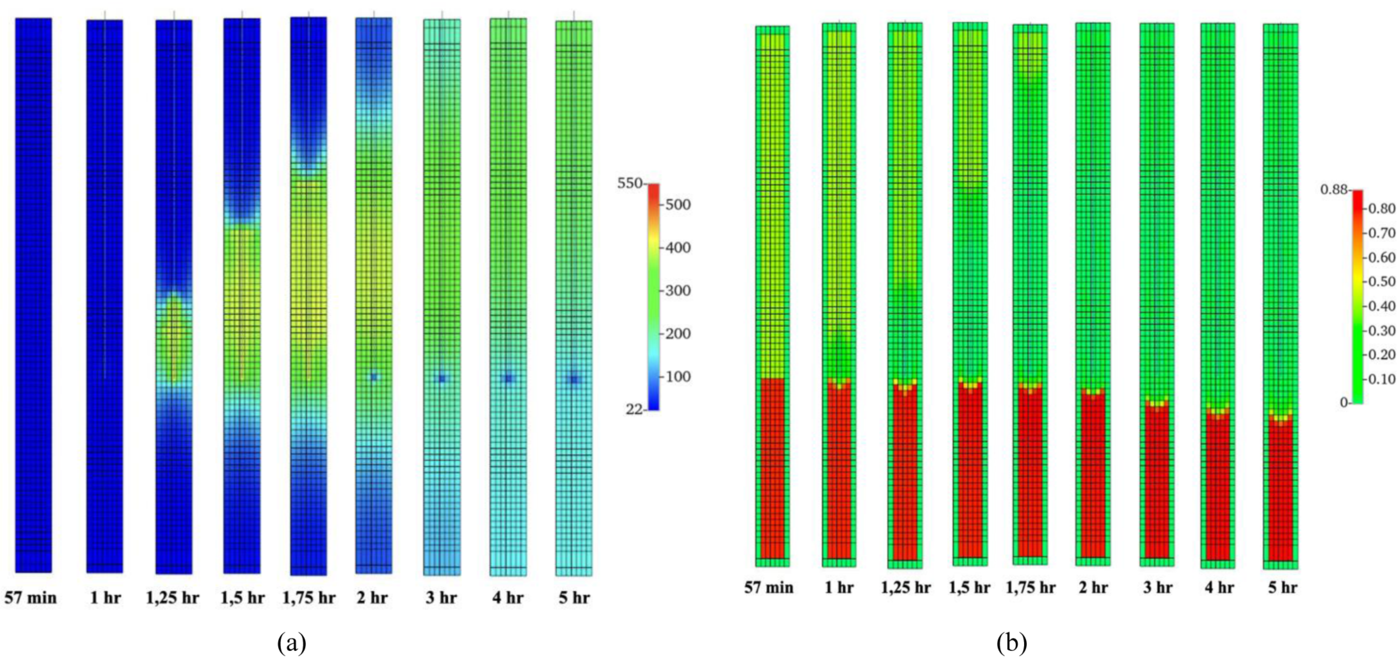


Figure 9—Temperature (a) and oil saturation (b) changes over time

To calibrate the oil oxidation processes, cumulative carbon dioxide volume was used as a key indicator. The model calibration results are presented in Figure 10. The final CO_2 values converged within a 2% margin of error (27,639 ml in the model versus 27,090 ml in the experiment), confirming the accuracy of the adopted kinetic parameters. Figure 10a shows that the gas production dynamics in the model differ from those observed experimentally. This discrepancy may be attributed to fluid displacement characteristics; in the experiment, variable injection rates likely caused fluids to exit at different rates, whereas the numerical model used a constant injection rate.

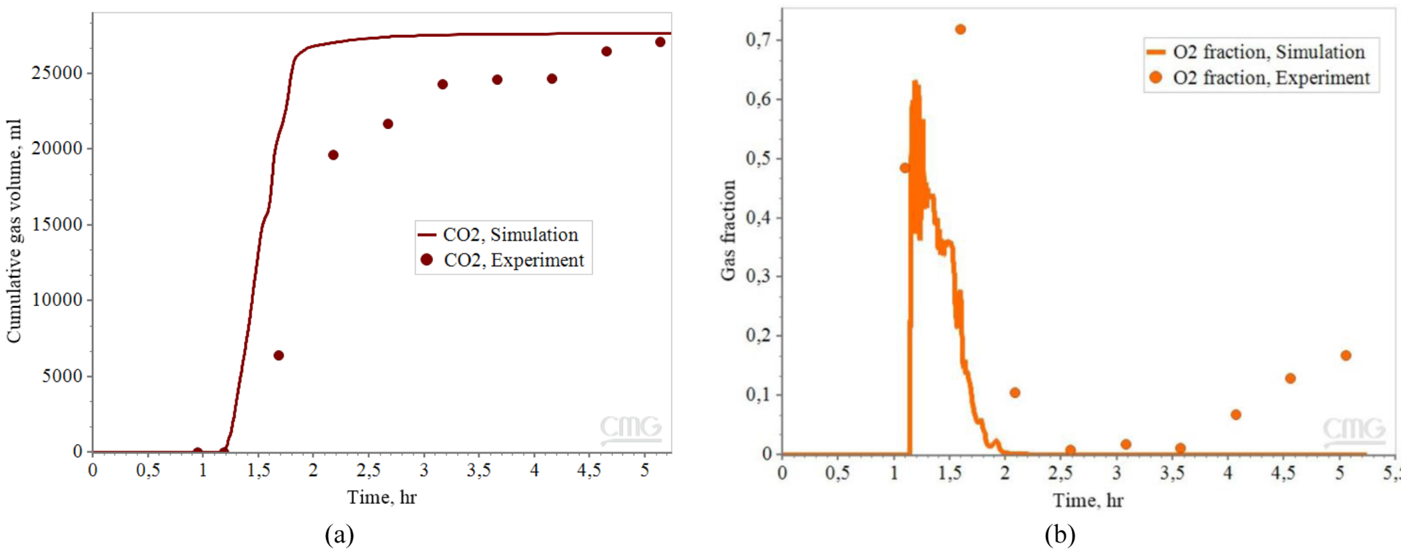


Figure 10—(a) Cumulative CO_2 volume; (b) Oxygen mole fraction

The results for the oxygen production dynamics are presented in Figure 10 (b). The data indicate that the main release of oxygen—corresponding to H_2O_2 decomposition—and its subsequent consumption exhibit similar trends in both the model and experiment during the time interval of 1 to 2.5 hours. The primary differences in gas dynamics occur after 3.5 hours, associated with operational procedures during the experiment, specifically switching the injection line from H_2O_2 to water, which led to discrepancies in the measured oxygen volume fraction.

Discussion

The experimental results confirmed the feasibility of using H_2O_2 injection combined with ISC for heavy oil recovery under reservoir conditions. The temperature profiles and combustion front velocities observed validate that peroxide decomposition efficiently initiates oxidation reactions, resulting in stable ISC propagation. The use of potassium permanganate as a catalyst effectively triggered peroxide decomposition, forming a solid-phase MnO_2 catalyst that ensures localized heat generation.

The observed recovery factor was high, reaching up to 81% in treated zones, driven by combined heating from peroxide decomposition and ISC. Oil properties varied significantly due to thermal and oxidative changes, including an increase in asphaltene content and manganese residues. This effect, while maintaining oil mobility due to high temperatures, suggests potential challenges for production facilities, such as increased solids and fines migration. The nearly complete oxygen utilization (98%) demonstrated the process efficiency; however, manganese displacement and partial clogging highlight the need for optimized catalyst concentration and injection strategies to prevent formation damage during field implementation.

The following results were summarized from the experiment:

1. A 6% potassium permanganate solution confirmed the efficiency of the previously proposed technology to initiate H_2O_2 decomposition and is recommended for field use.
2. The thermal effect observed in the laboratory study was combined. It includes the primary heating of the model with H_2O_2 decomposition and subsequent heating with the oxygen enriched ISC.
3. The observed ISC could be classified as wet or super-wet combustion with a water-to-air ratio of 1 L/m^3, average temperature of 320°C and with a fuel requirements of 21.8 kg/m^3 (12% of initial oil).
4. The velocity of the combustion front propagation was 0.33 m/h which is faster than air ISC, due to oxygen enrichment.

5. The recovery coefficient was 0.66 for the whole length model and approximately 0.81 for the zones with thermal treatment.
6. Oil properties varied widely, with density from 0.936 to 0.973 g/cm^3 and viscosity from 241 to 3074 mPa·s at 1 atm and 25°C, but high temperatures-maintained oil mobility; asphalt content increased due to oxidation and manganese residue. Additional findings include nearly complete oxygen utilization (98%), loss of initiator after first use, manganese displacement during filtration, presence of mechanical impurities causing potential clogging, and core damage during initiator injection that needs further investigation.

Numerical simulations showed good agreement with experimental results in predicting combustion front propagation and cumulative CO_2 production, confirming the kinetic model's applicability for field-scale scenarios. Incorporating peroxide decomposition into the reaction scheme improved accuracy by better representing the combined heating from peroxide and ISC.

However, analysis of Figure 8 reveals that while the model captures the timing and general thermal trends, it underestimates peak temperatures and overpredicts heat persistence in several zones. In Zone 8, the sharper simulated peak suggests the model assumes instantaneous peroxide decomposition, whereas experimental heating was more gradual, likely due to uneven catalyst distribution and diffusion limitations. Zone 7 shows close alignment, but faster experimental cooling indicates greater heat losses not fully captured in the model. The significant discrepancy in Zone 6, where the experimental peak reached ~500°C versus ~400°C in simulation, suggests local hot spot formation from intense oxidation near catalyst-rich areas, which homogenized model kinetics cannot capture. In Zones 4 and 3, experimental peaks were higher and more transient, implying that cumulative heat generation and oxygen availability sustained faster reactions than modeled.

Overall, while the simulations reliably predict combustion front advance, their underestimation of peak temperatures highlights limitations in representing reservoir heterogeneities, localized reaction intensification, and radial heat losses. Refining kinetic parameters and thermal boundary conditions will be crucial for enhancing model accuracy and ensuring effective field implementation of peroxide-based thermal EOR. Further research is needed to investigate enriched air injection kinetics through dedicated RTO experiments and to evaluate the long-term effects of catalyst residues on formation permeability and production equipment.

Conclusions

This study experimentally validated H_2O_2 injection combined with ISC as a promising thermal EOR method for heavy oil reservoirs. The laboratory experiment was conducted under reservoir conditions using consolidated sandstone core packs and dead oil samples from the target field at depths of 1650–1700 m. The experimental program replicated the full treatment sequence, including precursor injection, peroxide injection with ISC propagation, and final purging. Numerical simulations were developed and calibrated based on the experimental data, with model geometry reflecting the actual dimensions and properties of the core holder and target reservoir rock. The original kinetic model, consisting of four reactions, was extended by incorporating peroxide decomposition, and kinetic parameters were refined through calibration against experimental results. These updated models can be further integrated into sector development simulations for field-scale application of H_2O_2 based thermal EOR.

The key findings of this study are:

- **Catalyst effectiveness:** 6% potassium permanganate solution efficiently initiated H_2O_2 decomposition under reservoir conditions, confirming its practical applicability for field use.
- **Thermal and combustion performance:** The combined peroxide decomposition and oxygen-enriched ISC created effective reservoir heating. The combustion front propagated at 0.33 m/

h, faster than conventional air ISC, operating in a wet or super-wet regime with an average temperature of 320°C and fuel consumption of ~12% of initial oil.

- **Oil recovery:** The process achieved an overall recovery factor of 66% for the entire core model and up to 81% in thermally treated zones, significantly improving oil mobility despite increased asphaltene content and manganese residues.
- **Process efficiency and risks:** Nearly complete oxygen utilization (98%) was achieved, indicating high efficiency. However, manganese displacement and fines migration were observed, posing potential formation damage risks that require optimization of catalyst concentration and injection protocols.
- **Numerical model refinement:** Calibration against experimental results improved the model's predictive accuracy for thermal front propagation and CO_2 production. Nonetheless, the model underestimated peak temperatures, reflecting limitations in capturing localized reaction intensities and heat losses, which should be addressed in future kinetic studies.

Overall, this work demonstrates that H_2O_2 injection with catalytic decomposition is a feasible and effective alternative to conventional thermal recovery methods. To ensure safe and scalable field deployment, however, further research is required to better understand enriched air injection kinetics and the long-term effects of catalyst residues on reservoir performance. In particular, RTO experiments focusing on enriched air injection should be conducted to disentangle the individual impacts of partial oxidation, catalyst activity, and combustion gases, thereby refining process design and operational strategies for field applications.

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